

MAT1106 Design and Analysis of Experiments

Statistical Analysis of Data

Shamlan M. S. Reshamwala

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Which of the following statements conveys more information?

- A Total deaths due to COVID-19 in India during the period 01/01/2020 – 31/12/2020 were 1,48,738
- B Average deaths per month due to COVID-19 in India during the period 01/01/2020 – 31/12/2020 were 12,395

1. Statement A
2. Statement B
3. Both Statement A and Statement B
4. Neither Statement A nor Statement B

Deaths due to COVID-19 in India:

Jan	0
Feb	0
Mar	35
Apr	1119
May	4254
Jun	11,992
Jul	19,111
Aug	28,777
Sep	33,390
Oct	23,433
Nov	15,510
Dec	11,117

Calculated using data obtained from <https://ourworldindata.org/coronavirus/country/india?country=~IND>

The percentage of female students enrolled at various academic levels at the Institute of Chemical Technology (2020) is given below:

UG	32%
PG	46%
PhD	38%

What inferences can be drawn from this data?

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Measures of location

One type of measure useful for summarizing data defines the centre, or middle, of the sample. This type of measure is a measure of location.

- ▶ Mean (or average)
- ▶ Median
- ▶ Mode

Mean (or average)

The arithmetic mean is the sum of all the observations divided by the number of observations.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

The arithmetic mean is strongly influenced by extreme values.

For grouped (frequency type) data, the weighted arithmetic mean may be more appropriate, which is calculated as follows:

$$\frac{\sum_{i=1}^n x_i f_i}{\sum_{i=1}^n f_i} \quad (2)$$

Median

The median is the central value when all observations are arranged in order of magnitude. If there is an odd number of observations then it is simply the middle value; if there is an even number of observations then it is the average of the middle two.

The median is not generally influenced by extreme values.

The main weakness of the median is that it is determined mainly by the middle points in a sample and is less sensitive to the actual numeric values of the remaining data points.

Relation between mean and median

The relationship between the arithmetic mean and the median can be used to assess the symmetry of a distribution.

- ▶ For symmetric distributions the arithmetic mean is approximately the same as the median
- ▶ For positively skewed distributions, the arithmetic mean tends to be larger than the median
- ▶ For negatively skewed distributions, the arithmetic mean tends to be smaller than the median

Mode

The mode is the most frequently occurring value among all observations in a sample. When all values occur with equal frequency, mode will not exist.

A distribution with one mode is called unimodal; two modes, bimodal; three modes, trimodal; and so forth.

The mode is unaffected by the presence of extreme values.

Mode has no significance unless the number of observations is large.

Consider the following monthly salary levels (in INR) in a company:

Manager	50000
Salesman	15000
Cashier	8000
Foreman	7000
Two technicians, each	6000
Four workmen, each	4000

What is the average salary in the company? What is the median and mode?

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Shamlan M. S. Reshamwala

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Measures of spread

Measures of location alone cannot adequately represent or summarise statistical data because they do not provide any information about the spread or variability of the actual observations.

Two sets of data having the same measure of central tendency do not necessarily exhibit the same kind of dispersion or degree of heterogeneity in the data.

- ▶ Range
- ▶ Variance
- ▶ Standard deviation
- ▶ Coefficient of variation

Range

The range is the difference between the largest and smallest observations in a sample.

Advantage

- ▶ Very easy to compute once the sample points are ordered

Disadvantages

- ▶ Based on only two of the observations and may not be representative of the whole dataset
- ▶ Gives no information regarding how the data are distributed between the two extremes
- ▶ Depends on the sample size n ; the larger n is, the larger the range tends to be
- ▶ Difficult to compare ranges from data sets of differing size

Variance

The variance (denoted by s^2) is a measure of the degree to which individual observations in a dataset deviate from the mean value, i.e., it is the average deviation from the mean across all observations.

It is calculated by squaring the difference of each individual observation from the mean (squared to remove any negative differences), adding them together, and dividing by the total number of observations.

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \quad (1)$$

The more usual form for this measure is with $n - 1$ in the denominator rather than n . This is because the latter tends to underestimate variance by a factor of $(n - 1)/n$. This factor is considerable for small samples but tends to be negligible for large samples.

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \quad (2)$$

Standard deviation

Standard deviation (denoted by s or sd) is the square root of the variance.

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}} \quad (3)$$

Standard deviation for grouped frequency distribution is given by

$$s = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n - 1}} \quad (4)$$

If individual observations are all close to the mean, the standard deviation will be small.

If all observations were equal to the mean, the standard deviation would be zero.

If the observations vary widely, the standard deviation will be substantially larger.

Coefficient of variation

The mean and standard deviation are the most widely used measures of location and spread. The coefficient of variation (CV) is used to relate the arithmetic mean and the standard deviation to each other.

$$CV = 100\% \times (s/\bar{x}) \quad (5)$$

The coefficient of variation is useful because the standard deviation of data must always be understood in the context of the mean of the data.

For comparison between data sets with different units or widely different means, the coefficient of variation should be used instead of the standard deviation. This is because while the latter is an absolute measure of spread, the former is a relative measure.

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Shamlan M. S. Reshamwala

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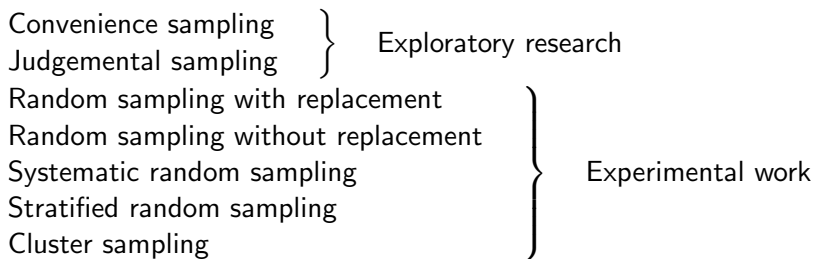
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Studying populations

Populations often are too large to be adequately studied in a specified time period or within designated budgetary constraints.

If it is not feasible to collect information on every item in a population, inferences on the population can be made by studying a representative subset of the data, i.e., a sample.

How is sampling carried out?



Regardless of which sampling technique is used, the sample should be representative of the population under study.

Inferring population parameters from sample estimates

Consider a sample of data $x_1, \dots, x_n$, where x_1 corresponds to the first sample point and x_n corresponds to the n th sample point.

Presuming that the sample is drawn from some population P , what inferences or conclusions can be made about P from the sample?

Parameters and statistics

A parameter is a numerical characteristic of a population.

A statistic is a numerical characteristic that is computed from a sample of observations.

Parameters are often denoted by Greek letters to reinforce the notion that they are (generally) unknown.

Ten random samples were drawn from an infinite population of numbers 0–9, and sample means and variances were calculated:

Sample	Value	Sample mean	Sample variance
1	2,4,1,3,0	2.0	2.5
2	4,2,1,6,7	4.0	6.4
3	9,6,0,3,1	3.8	13.7
4	1,0,4,8,0	2.6	11.8
5	2,2,3,6,8	4.2	7.2
6	3,7,5,7,0	4.4	8.8
7	9,9,5,6,2	6.2	8.7
8	8,9,5,7,9	7.6	2.8
9	8,5,4,7,5	5.8	2.7
10	7,7,9,2,1	5.2	12.2
Means		4.58	7.69

In the infinite population of numbers 0–9, the proportion of each number is equal to $1/10$. This equal proportion permits us to evaluate the population mean (μ) and variance (σ^2) exactly:

$$\mu = \frac{0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9}{10} = 4.5$$

$$\sigma^2 = \frac{(0 - 4.5)^2 + (1 - 4.5)^2 + \dots + (9 - 4.5)^2}{10} = 8.25$$

These values compare favourably with the means of the ten group-means (4.58) and variances (7.69).

Distribution of values

The pattern of values obtained when a quantity is measured in large numbers is called a distribution.

Distributions can be discrete or continuous.

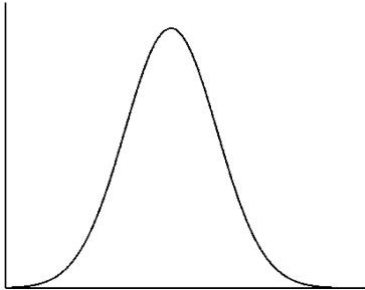
The normal distribution

Quantitative data follow a wide range of distributions.

The most common distribution is symmetrical and unimodal, with a single peak in the middle and equal tails at either side.

This distinctive bell-shaped distribution is known as 'normal' or 'Gaussian'.

Many random variables tend to follow approximately a normal distribution.

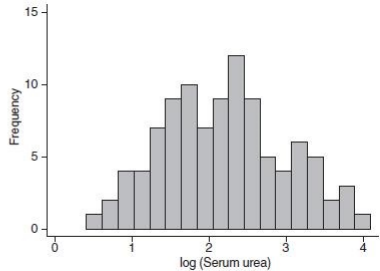
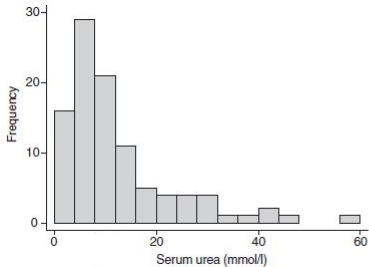


The normal distribution

Transformation of data

Skewed distributions have a substantially longer tail on one side. Such data can be transformed to obtain a more symmetrical (normal) distribution, a crucial requirement for many statistical techniques (such as t tests, regression and analysis of variance).

Logarithmic transformation is most commonly used with positively skewed data. Calculations and statistical tests can be carried out on the transformed data before converting results back to the original scale.



Histogram of serum urea levels from 100 patients (left); the same data logarithmically transformed (right)

The mean calculated in such a manner is called the geometric mean and is measured in the same units as the raw data.

However, the backtransformed standard deviation will not have the same units, as calculation of the standard deviation of the log transformed data requires taking the difference between each log observation and the log geometric mean. The difference between the log of two numbers is the log of their ratio. As a ratio is a dimensionless pure number, the standard deviation cannot be transformed back to the original scale.

For positively skewed data, the geometric mean will be smaller than the arithmetic mean and will be more representative of the data.

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Shamlan M. S. Reshamwala

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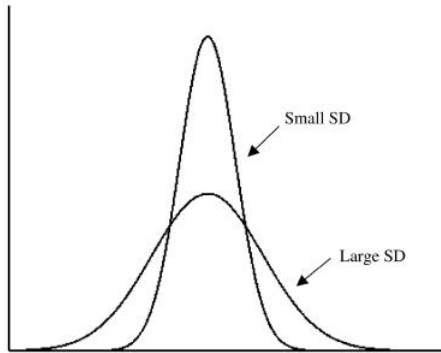


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Standard normal distribution

Normal distributions do not necessarily have the same means and standard deviations.

A normal distribution with a mean of 0 and standard deviation of 1 is called a standard normal deviation.



Normal curves with small and large SDs but same means

The z score

A value from any normal distribution can be transformed into its corresponding value on a standard normal distribution using the following formula:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where z is the value on the standard normal distribution, x is the value on the original distribution, μ is the mean of the original distribution, and σ is the standard deviation of the original distribution. Note that the z score can be only defined if the population parameters are known.

The z score is the distance from the sample mean to the population mean in units of the standard deviation.

The probability of observing a standard normal value below or above the z score can be determined using a table of the standard normal distribution.

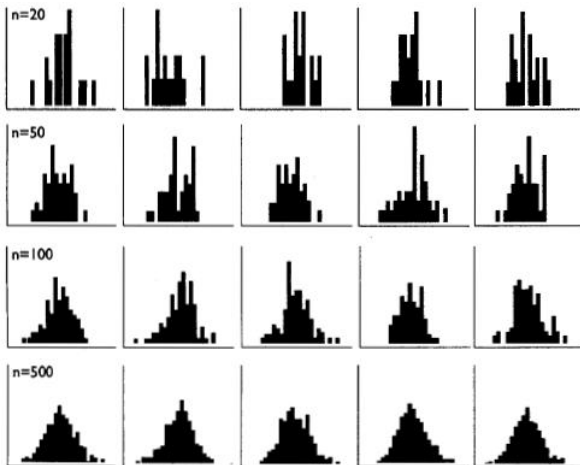
The central limit theorem

Samples from a population in which the true distribution is normal will not necessarily look normal themselves, especially if the sample is small.

Just as separate samples selected at random from the same population differ, so will calculated values like the mean.

The central limit theorem states that means from random samples (>30 in practice) from any distribution will themselves have a normal distribution.

Consequently, the distribution of the data can often be ignored if the sample size is large.



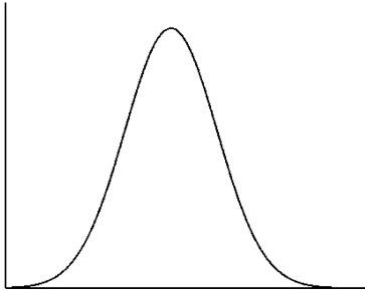
Random samples from normal distributions — five samples of size 20, 50, 100 and 500.

Features of the normal distribution

As normal distributions are symmetrical around their mean, the mean, median and mode are all equal. The area under the normal curve is equal to 1.

The normal distribution is entirely defined by two quantities: its mean and its standard deviation (SD).

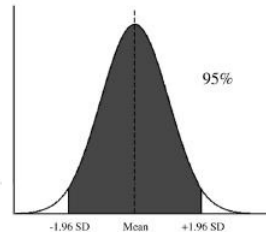
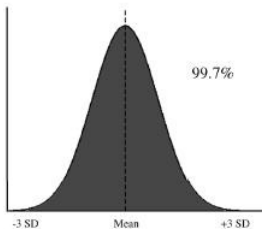
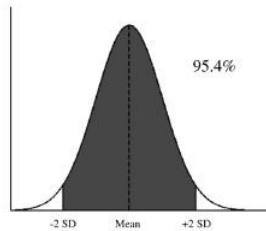
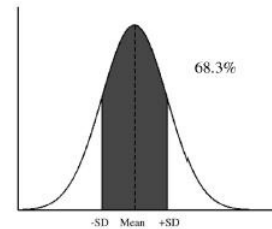
The mean determines where the peak occurs and the SD determines the shape of the curve.



The normal distribution

Because the normal curve is defined entirely by its mean and SD, the following rules will always apply regardless of the specific values of these quantities:

- ▶ 68.3% of the distribution falls within 1 SD of the mean (i.e., between mean $-$ SD and mean $+$ SD)
- ▶ 95.4% of the distribution falls between mean $-$ 2 SD and mean $+$ 2 SD
- ▶ 99.7% of the distribution falls between mean $-$ 3 SD and mean $+$ 3 SD
- ▶ 95% of the distribution falls between mean $-$ 1.96 SD and mean $+$ 1.96 SD



Areas under the normal curve

Reference range

The reference range is another useful measure of variability in a dataset.

The 95% reference range can be calculated as (mean – 1.96 SD) to (mean + 1.96 SD), provided that the data are (approximately) normally distributed. This range will contain approximately 95% of the data.

It is also possible to define a 90% reference range, a 99% reference range and so on in the same way, but conventionally the 95% reference range is the most commonly used.

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Shamlan M. S. Reshamwala

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Standard error

As a consequence of the central limit theorem, the distribution of sample means is known to be normal, regardless of the underlying distribution of the population. Because of this it is possible to calculate the mean and SD of the sample means.

The mean of all the sample means is equal to the population mean, while the SD is reduced by a factor of $\sqrt{n}$ from that of the original population. In order to distinguish it from the sample SD, it is known as the standard error (SE).

$$SE = \sigma / \sqrt{n} \quad (1)$$

where σ is the SD of the population and n is the sample size.

In practice, σ is unknown but the sample SD will generally provide a good estimate and so the SE is estimated by the following equation:

$$SE = s/\sqrt{n} \quad (2)$$

where s is the sample SD.

It can be seen from this that the SE will always be considerably smaller than the SD in a sample. This is because there is less variability between the sample means than between individual values.

It is also clear that larger sample sizes lead to smaller standard errors (because the denominator, $\sqrt{n}$, is larger). In other words, large sample sizes produce more precise estimates of the population value in question, as the extent of chance variation is reduced.

If the population mean (μ) is 35.4 and the SD (σ) is 2.5, what will be the mean and SE of the distribution of means of sample size 4 from the same population?

$$\text{Mean} = 35.4$$

$$\text{SE} = 2.5/\sqrt{4} = 1.25$$

Confidence interval

Because sample means are normally distributed, it should be possible to calculate a range of values in which 95% of sample means lie.

In practice, the population mean (the mean of all sample means) is unknown.

The 95% confidence interval provides a range of values within which the true population mean is likely to lie. The 95% confidence interval is calculated from a single sample using the mean and SE.

It is defined as follows: (sample mean $-$ 1.96 SE) to (sample mean $+$ 1.96 SE).

Using a 99% confidence interval (calculated as $\text{mean} \pm 2.56 \text{ SE}$) will make it more likely that the true value is contained within the range. However, the cost of this change is that the range will be wider and therefore less precise.

While the SD and 95% reference range describe variability within a sample, the SE and 95% confidence interval describe variability between samples, and therefore provide a measure of the precision of a population value estimated from a single sample.

In other words, a 95% confidence interval provides a range of values within which the true population value of interest is likely to lie.

Confidence intervals for smaller samples

The calculation of a 95% confidence interval relies on two assumptions:

- ▶ the distribution of sample means is approximately normal
- ▶ the population SD can be approximated by the sample SD

These assumptions will generally be valid if the sample is sufficiently large. However, there may be occasions when these assumptions break down.

If the population distribution is extremely non-normal and the sample size is very small, it may be appropriate to use Student's t distribution.

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Shamlan M. S. Reshamwala

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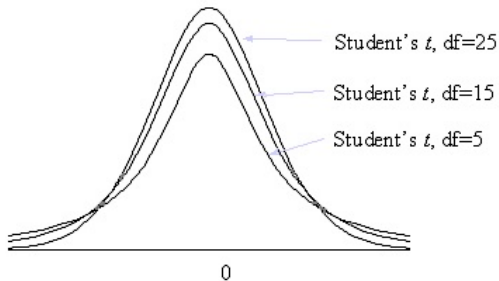
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Student's t distribution

The Student's t distribution (or simply the t distribution) is similar in shape to the normal distribution, being symmetrical and unimodal, but is generally more spread out with longer tails.

The exact shape depends on a quantity known as the 'degrees of freedom', which refers to the number of independent observations in a set of data.

When estimating a mean score or a proportion from a single sample, the number of independent observations is equal to the sample size minus one. Hence, the distribution of the t statistic from samples of size 8 would be described by a t distribution having 7 degrees of freedom.



Student's t distributions

Thus, Student's t distribution is not a unique distribution but is a family of distributions indexed by the degrees of freedom.

The t distribution tends toward the normal distribution (i.e., it becomes less spread out) as the degrees of freedom/sample sizes increase.

Calculating a confidence interval using the t distribution

Calculating a confidence interval using the t distribution is very similar to calculating it using the normal distribution.

In the case of the normal distribution, the calculation is based on the fact that 95% of sample means fall within 1.96 SEs of the population mean. The longer tails of the t distribution mean that it is necessary to go slightly further away from the mean to pick up 95% of all sample means.

The alternative multiplication factor depends on the degrees of freedom of the t distribution in question, and some typical values are presented in the following table.

Multiplication factors for 95% confidence intervals based on the t distribution

Sample size	10	20	30	40	50	200
Multiplication factor	2.26	2.09	2.05	2.02	2.01	1.97

As the sample sizes increase, the multiplication factors decrease toward 1.96 (the multiplication factor for an infinite sample size is 1.96).

The larger multiplication factors for smaller samples result in a wider confidence interval, and this reflects the uncertainty in the estimate of the population SD by the sample SD.

The use of the t distribution is known to be extremely robust and will therefore provide a valid confidence interval unless the population distribution is severely non-normal.

Student's t statistic

The t statistic is a ratio of the departure of an estimated parameter from its notional value and its standard error. It is calculated as follows:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \quad (1)$$

If the population parameters are known, then rather than computing the t statistic, the z score can be computed; analogously, rather than using a t test, the z test is used.

The binomial distribution

Probability distributions for a specific outcome (called a 'success', p) arising from repeated independent events (or trials, n) for which there are just two possible outcomes with fixed probabilities summing to one are called binomial distributions.

A flip of an unbiased coin has two possible outcomes: head or tail. Successive flips of such a coin will lead to outcomes that are binomially distributed.

The Poisson distribution

When the number of trials is very large (tending to infinity, $n \rightarrow \infty$), and the probability of a specific outcome is very small (tending to zero, $p \rightarrow 0$), the probability distribution of such an outcome will follow a Poisson distribution.

Examples of random variables that have a Poisson distribution:

- ▶ number of accidents at a road crossing
- ▶ number of bacterial colonies per unit area of an agar plate
- ▶ number of suicides in a year in a given area

For large n , the binomial and Poisson distributions (both of which are discrete probability distributions) can be approximated very closely by a normal distribution (which is a continuous probability distribution).

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Shamlan M. S. Reshamwala

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Hypothesis testing

Hypothesis testing is a statistical procedure for testing whether chance is a plausible explanation of an experimental finding.

Hypothesis testing is generally used when some comparison is to be made. It provides an objective framework for making decisions using probabilistic methods, rather than relying on subjective impressions.

Tests of significance are applied after setting up a hypothesis. The choice of which statistical test to use depends on the format of the data and the study design.

Null and alternate hypotheses

The null hypothesis, denoted by H_0 , is the hypothesis that is to be tested.

The alternate hypothesis, denoted by H_1 , is the hypothesis that in some sense contradicts the null hypothesis.

For example, if the null hypothesis is that the mean of a population under study (say, μ) is equal to the mean of the general population (μ_0), the alternate hypothesis could be that the former is not equal to the latter.

$H_0 : \mu = \mu_0$ (null hypothesis)

$H_1 : \mu \neq \mu_0$ (two-tailed alternate hypothesis)

$H_1 : \mu > \mu_0$ (right-tailed alternate hypothesis)

$H_1 : \mu < \mu_0$ (left-tailed alternate hypothesis)

The setting up of the alternate hypothesis is very important to decide whether a one-tailed (right or left) or two-tailed test is to be used.

One- and two-tailed tests

A one-tailed test is a test in which the values of the parameter being studied (in this case μ) under the alternate hypothesis are allowed to be either greater than or less than the values of the parameter under the null hypothesis (μ_0), but not both.

A two-tailed test is a test in which the values of the parameter being studied under the alternate hypothesis are allowed to be either greater than or less than the values of the parameter under the null hypothesis.

Possible outcomes of hypothesis testing

Suppose the only possible decisions are whether H_0 is true or H_1 is true. If we decide H_0 is true, then we say we accept H_0 . If we decide H_1 is true, then we state that H_0 is not true or, equivalently, that we reject H_0 .

Thus, four possible outcomes can occur:

1. We accept H_0 , and H_0 is in fact true.
2. We accept H_0 , and H_1 is in fact true.
3. We reject H_0 , and H_0 is in fact true.
4. We reject H_0 , and H_1 is in fact true.

In actual practice, it is impossible, using hypothesis-testing methods, to prove that the null hypothesis is true. Thus, in particular, if we accept H_0 , then we have actually failed to reject H_0 .

If H_0 is true and H_0 is accepted, or if H_1 is true and H_0 is rejected, then the correct decision has been made. If H_0 is true and H_0 is rejected, or if H_1 is true and H_0 is accepted, then an error has been made. The two types of errors are generally treated differently.

Type I and type II errors

The probability of a type I error is the probability of rejecting the null hypothesis when H_0 is true.

The probability of a type II error is the probability of accepting the null hypothesis when H_1 is true. If we write

$$P(\text{rejecting } H_0 \text{ when } H_0 \text{ is true}) = P(\text{reject } H_0 | H_0) = \alpha$$

$$P(\text{accepting } H_0 \text{ when } H_1 \text{ is true}) = P(\text{accept } H_0 | H_1) = \beta$$

then α and β are the sizes of type I and type II errors, respectively.

The probability of a type I error is commonly referred to as the significance level of a test.

The probability of a type II error is used to measure the power of a test in detecting the falsity of the null hypothesis. The power of a test is given by $1 - \beta = P(\text{reject } H_0 | H_1)$.

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Shamlan M. S. Reshamwala

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Minimising errors

It is desirable that the test procedure be so framed that both types of error are minimised.

However, this is not possible, because an attempt to reduce one type of error is generally accompanied by an increase in the other type.

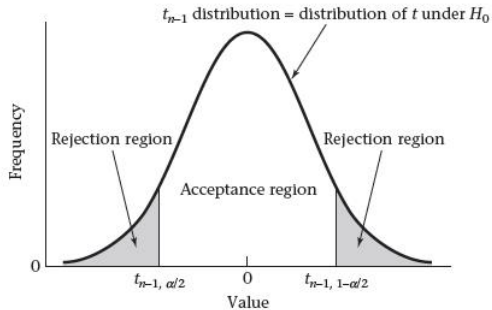
- ▶ As α decreases, β increases
- ▶ As α increases, β decreases

The general strategy is designed so as to limit α to a specified value (usually 5% or 1%) and at the same time to minimise β .

Critical region and level of significance

The level of significance of a test is based on the sample mean ($\bar{x}$). If $\bar{x}$ is sufficiently smaller (or greater) than μ_0 , then H_0 is rejected; otherwise, H_0 is accepted.

This test is reasonable because if H_0 is true, then the most likely values of $\bar{x}$ tend to cluster around μ_0 , whereas if H_1 is true, the most likely values of $\bar{x}$ tend to cluster around μ .



Rejection and acceptance regions of the t statistic
under the null hypothesis

A region (corresponding to a t statistic) in the sample space S that amounts to rejection of H_0 is termed the 'critical (or rejection) region'.

If ω is the critical region and if $t = t(x_1, \dots, x_n)$ is the value of the statistic based on a random sample of size n , then $P[t \in \omega | H_0] = \alpha$ and $P[t \in \omega' | H_1] = \beta$, where ω' the complementary set of ω , is called the 'acceptance region'. Note that $\omega \cup \omega' = S$.

The probability of α , that is, the probability that a random sample of the t statistic belongs to the critical region is known as the 'level of significance'. In other words, the level of significance is the size of the type I error. As mentioned before, the levels of significance usually employed in testing of hypothesis are 5% and 1%.

Critical value or significant value

The value of the test statistic which separates the critical (or rejection) region and the acceptance region is called the 'critical value' or 'significant value'. It depends on the level of significance used and the alternate hypothesis, whether it is two-tailed or one-tailed.

The critical region is denoted by

- ▶ $t_{n-1,\alpha}$ for left-tailed
- ▶ $t_{n-1,1-\alpha}$ for right-tailed
- ▶ $t_{n-1,\alpha/2}$ for two-tailed

alternate hypotheses.

The actual α level used should depend on the relative importance of type I and type II errors because the smaller α is made for a fixed sample size (n), the larger β becomes.

Traditionally, an α level of exactly 0.05 is used most frequently.

Procedure for testing of hypotheses

The steps followed in testing a statistical hypothesis are summarised below:

1. Set up the null hypothesis (H_0).
2. Set up the alternate hypothesis (H_1).
3. Choose the appropriate level of significance (α).
4. Compute the test statistic (t or z).
5. Compare the computed value of the test statistic with the significant value at the chosen level of significance.

One-sample t test for the mean of a normal distribution with unknown variance (left-tailed test; alternate mean $<$ null mean)

To test the hypothesis $H_0 : \mu = \mu_0$, σ unknown versus $H_1 : \mu < \mu_0$, σ unknown with a significance level of α , compute

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad (1)$$

If $t < t_{n-1,\alpha}$, H_0 is rejected.

If $t \geq t_{n-1,\alpha}$, H_0 is accepted.

One-sample z test for the mean of a normal distribution with known variance (left-tailed test; alternate mean $<$ null mean)

To test the hypothesis $H_0 : \mu = \mu_0$, σ known versus $H_1 : \mu < \mu_0$, σ known with a significance level of α , compute

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \quad (2)$$

If $z < z_\alpha$, H_0 is rejected.

If $z \geq z_\alpha$, H_0 is accepted.

One-sample t test for the mean of a normal distribution with unknown variance (right-tailed test; alternate mean $>$ null mean)

To test the hypothesis $H_0 : \mu = \mu_0$, σ unknown versus $H_1 : \mu > \mu_0$, σ unknown with a significance level of α , compute

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad (3)$$

If $t > t_{n-1, 1-\alpha}$, H_0 is rejected.

If $t \leq t_{n-1, 1-\alpha}$, H_0 is accepted.

One-sample z test for the mean of a normal distribution with known variance (right-tailed test; alternate mean $>$ null mean)

To test the hypothesis $H_0 : \mu = \mu_0$, σ known versus $H_1 : \mu > \mu_0$, σ known with a significance level of α , compute

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \quad (4)$$

If $z > z_{1-\alpha}$, H_0 is rejected.

If $z \leq z_{1-\alpha}$, H_0 is accepted.

One-sample t test for the mean of a normal distribution with unknown variance (two-tailed test)

To test the hypothesis $H_0 : \mu = \mu_0$, σ unknown versus $H_1 : \mu \neq \mu_0$, σ unknown with a significance level of α , compute

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad (5)$$

If $t < t_{n-1, \alpha/2}$ or $t > t_{n-1, 1-\alpha/2}$, H_0 is rejected.

If $t \geq t_{n-1, \alpha/2}$ or $t \leq t_{n-1, 1-\alpha/2}$, H_0 is accepted.

One-sample z test for the mean of a normal distribution with known variance (two-tailed test)

To test the hypothesis $H_0 : \mu = \mu_0$, σ known versus $H_1 : \mu \neq \mu_0$, σ known with a significance level of α , compute

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \quad (6)$$

If $z < z_{\alpha/2}$ or $z > z_{1-\alpha/2}$, H_0 is rejected.

If $z \geq z_{\alpha/2}$ or $z \leq z_{1-\alpha/2}$, H_0 is accepted.

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Shamlan M. S. Reshamwala

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The p value

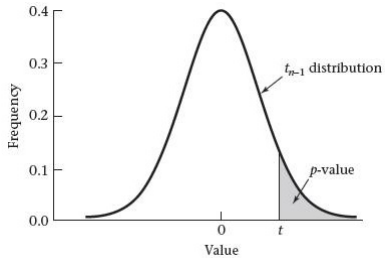
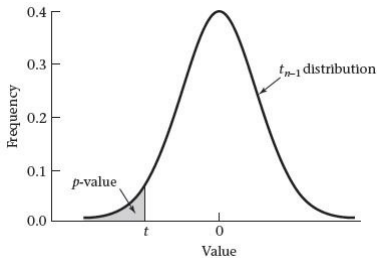
Performing significance tests at different α levels and accepting or rejecting H_0 in each instance is tedious.

Instead, significance tests can be effectively performed at all α levels by obtaining the p value for the test.

The p value measures how likely it is that any observed difference between groups is due to chance. In other words, the p value is the probability of seeing the observed difference, or greater, just by chance if the null hypothesis is true.

Being a probability, p can take any value between 0 and 1.

Values close to 0 indicate that the observed difference is unlikely to be due to chance, whereas a p value close to 1 suggests there is no difference between groups other than that due to random variation.



p values for one-tailed t tests

Under the null hypothesis, the t statistic follows a t_{n-1} distribution, while the z statistic follows a normal distribution.

For a left-tailed test, the p value is given by

$$p = P(t_{n-1} \leq t) \quad \text{or} \quad p = P(Z \leq z) \quad (1)$$

For a right-tailed test, the p value is given by

$$p = P(t_{n-1} \geq t) \quad \text{or} \quad p = P(Z \geq z) \quad (2)$$

The p value for a two-tailed test is given by

$$\begin{aligned} p &= \begin{cases} 2 \times P(t_{n-1} \leq t) & \text{if } t \leq 0 \\ 2 \times [1 - P(t_{n-1} \leq t)] & \text{if } t > 0 \end{cases} \quad \text{or} \\ &= \begin{cases} 2 \times P(Z \leq z) & \text{if } z \leq 0 \\ 2 \times [1 - P(Z \leq z)] & \text{if } z > 0 \end{cases} \end{aligned} \quad (3)$$

Thus, if $t \leq 0$, then $p = 2$ times the area under a t_{n-1} distribution to the left of t ; if $t > 0$, then $p = 2$ times the area under a t_{n-1} distribution to the right of t .

The p value is important because it tells us exactly how significant our results are without performing repeated significance tests at different α levels. How small should the p value be for results to be considered statistically significant? Some commonly followed criteria are:

If $0.01 \leq p < 0.05$, then the results are significant.

If $0.001 \leq p < 0.01$, then the results are highly significant.

If $p < 0.001$, then the results are very highly significant.

If $p > 0.05$, then the results are considered not statistically significant.

However, if $0.05 \leq p < 0.10$, then a trend toward statistical significance is sometimes noted.

Relationship between hypothesis testing and confidence intervals

If $H_0 : \mu = \mu_0$, and $H_1 : \mu \neq \mu_0$, H_0 is rejected with a two-tailed level of significance α if and only if the confidence interval at the same level of significance for μ does not contain μ_0 .

Conversely, H_0 is accepted with a two-tailed level of significance α if and only if the confidence interval at the same level of significance for μ does contain μ_0 .

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Shamlan M. S. Reshamwala

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Power of a test

The probability of a type II error (β) is the probability of accepting the null hypothesis when H_1 is true. The probability of a type II error is used to measure the power of a test.

Power is defined as the probability of correctly rejecting a false null hypothesis, or the probability of not committing a type II error.

The power of a test is given by $1 - \beta = P(\text{reject } H_0 | H_1)$.

Although there are no formal standards for power, the conventional value for β of a statistical test is 0.2, the power being equal to 0.8.

Factors affecting the power of a test

Several factors affect the power of a statistical test:

Sample size — larger the sample size, higher the power.

Standard deviation — smaller the standard deviation, higher the power.

Difference between hypothesised and true mean — larger the difference, higher the power.

Significance level — more stringent the significance level, lower the power.

One- versus two-tailed tests — power is higher with a one-tailed test than with a two-tailed test as long as the hypothesized direction is correct. A one-tailed test at the 0.05 level has the same power as a two-tailed test at the 0.10 level.

Sample-size determination

A research study may be conducted to estimate some quantity, say x , in a specified population with a given degree of precision. It is essential that the study be of an appropriate size to achieve its aims.

The problem of determining sample size can be summarized as follows: given that a one-tailed significance test will be conducted at level α and that the true alternate mean is expected to be μ_1 , what sample size is needed to be able to detect a significant difference with probability $1 - \beta$?

Sample-size estimation when testing for the mean of a normal distribution (one-tailed alternate)

Suppose we wish to test $H_0 : \mu = \mu_0$ versus $H_1 : \mu = \mu_1$ where the data are normally distributed with mean μ and known variance σ^2 . The sample size needed to conduct a one-tailed test with significance level α and probability of detecting a significant difference $= 1 - \beta$ is

$$n = \frac{\sigma^2(z_{1-\beta} + z_{1-\alpha})^2}{(\mu_0 - \mu_1)^2} \quad (1)$$

Notice that the sample size is very sensitive to the alternate mean chosen.

The sample size is inversely proportional to $(\mu_0 - \mu_1)^2$. Thus, if the absolute value of the distance between the null and alternate means is halved, then the sample size needed is four times as large.

Similarly, if the distance between the null and alternate means is doubled, then the sample size needed is $1/4$ as large.

Sample-size estimation when testing for the mean of a normal distribution (two-tailed alternate)

Suppose we wish to test $H_0 : \mu = \mu_0$ versus $H_1 : \mu = \mu_1$ where the data are normally distributed with mean μ and known variance σ^2 . The sample size needed to conduct a two-tailed test with significance level α and probability of detecting a significant difference $= 1 - \beta$ is

$$n = \frac{\sigma^2(z_{1-\beta} + z_{1-\alpha/2})^2}{(\mu_0 - \mu_1)^2} \quad (2)$$

Note that this sample size is always larger than the corresponding sample size for a one-tailed test because $z_{1-\alpha/2}$ is larger than $z_{1-\alpha}$.

One question that arises is how to estimate the parameters necessary to compute sample size. It usually is easy to specify the magnitude of the null mean (μ_0). Similarly, by convention the type I error (α) is usually set at 0.05 and power ($1 - \beta$) at 0.80. The appropriate values for μ_1 and σ^2 are usually unknown, but may be obtained from previous work, similar experiments, or prior knowledge of the underlying distribution. In the absence of such information, the parameter μ_1 is sometimes estimated by assessing what a scientifically important difference $|\mu_0 - \mu_1|$ would be in the context of the problem being studied. A small pilot study may be conducted to obtain estimates of μ_1 and σ^2 for the purpose of estimating the sample size needed to conduct the major investigation.

Factors affecting the sample size

Several factors affect the sample size:

The sample size increases as σ^2 increases.

The sample size increases as the significance level is made smaller (α decreases).

The sample size increases as the required power increases ($1 - \beta$ increases).

The sample size decreases as the absolute value of the distance between the null and alternate means ($|\mu_0 - \mu_1|$) increases.

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Shamlan M. S. Reshamwala

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Two-sample tests

The tests discussed till now were one-sample tests, in which the underlying parameters of the population from which the sample was drawn were compared with comparable values from other generally large populations whose parameters were assumed to be known.

A more frequently encountered situation is the two-sample hypothesis-testing problem. In a two-sample hypothesis-testing problem, the underlying parameters of two different populations, neither of whose values is assumed known, are compared.

Paired t test

Two samples are said to be paired when each data point in the first sample is matched and is related to a unique data point in the second sample.

If the difference between the paired data is Δ , the null hypothesis can be stated as $H_0 : \Delta = 0$, while the alternate hypothesis will be $H_1 : \Delta \neq 0$. The t statistic for the paired t test is computed as follows:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \quad (1)$$

where $\bar{d}$ is the mean of the differences, s_d is the standard deviation of the differences, and n is the number of matched pairs.

s_d is computed as follows:

$$s_d = \sqrt{\frac{\sum_{i=1}^n (d_i - \bar{d})^2}{n - 1}} \quad (2)$$

Unpaired t test

The most common comparison is probably that of two means arising from unpaired data (i.e., comparison of data from two independent groups). If the means of the two sets of data are μ_1 and μ_2 , the null hypothesis is $H_0 : \mu_1 = \mu_2$, while the alternate hypothesis is $H_1 : \mu_1 \neq \mu_2$. The t statistic for the unpaired t test is computed as follows:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad (3)$$

where n_1 and n_2 are the two sample sizes.

s in this instance is the pooled standard deviation of the two samples, and is computed as follows:

$$\begin{aligned} s &= \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)}} \\ &= \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)}} \end{aligned} \tag{4}$$

In the paired case attention is focused on the mean of the differences; in the unpaired case interest is in the difference of the means.

Because the sample sizes in the unpaired case may be (and indeed usually are) different, the combined standard error takes this into account and gives more weight to the larger sample size because this is likely to be more reliable.

Assumptions and limitations of the t tests

- ▶ The one sample t test requires that the data have an approximately normal distribution
- ▶ The paired t test requires that the distribution of the differences are approximately normal
- ▶ The unpaired t test relies on the assumption that the data from the two samples are both normally distributed, and that the standard deviations from the two samples are approximately equal

If assumptions of normality are violated, appropriate transformation of the data may be used before performing any calculations.

Transformations may also be useful if the standard deviations are very different in the unpaired case.

It may not always be possible to get around these limitations; where this is the case, alternative tests can be used.

t tests are restricted to cases where comparison is to be made between one or two groups. To explore differences in means across three or more groups, analysis of variance (ANOVA) can be employed.

Table of t values

Tail Probability		Degrees of Freedom																			Confidence Levels	
1-tail	2-tail	1	2	3	4	5	6	7	8	9	10	12	15	20	25	30	40	50	1 00	1000 ≈ ∞	2-sided	1-sided
0.5	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	80%	
0.4	0.8	0.325	0.289	0.277	0.271	0.267	0.265	0.263	0.262	0.261	0.260	0.259	0.258	0.257	0.256	0.256	0.255	0.255	0.254	0.253		
0.3	0.6	0.727	0.617	0.584	0.569	0.559	0.553	0.549	0.546	0.543	0.542	0.539	0.536	0.533	0.531	0.530	0.529	0.528	0.526	0.525		
0.25	0.5	1.000	0.816	0.765	0.741	0.727	0.718	0.711	0.706	0.703	0.700	0.695	0.691	0.687	0.684	0.683	0.681	0.679	0.677	0.675		
0.2	0.4	1.376	1.061	0.978	0.941	0.920	0.906	0.896	0.889	0.883	0.879	0.873	0.866	0.860	0.856	0.854	0.851	0.849	0.845	0.842		
0.17	0.34	1.691	1.242	1.132	1.082	1.054	1.036	1.024	1.015	1.008	1.002	0.994	0.986	0.977	0.973	0.970	0.966	0.963	0.959	0.955	80%	
0.15	0.3	1.963	1.386	1.250	1.190	1.156	1.134	1.119	1.108	1.100	1.093	1.083	1.074	1.064	1.058	1.055	1.050	1.047	1.042	1.037		
0.14	0.28	2.125	1.467	1.315	1.248	1.211	1.187	1.171	1.159	1.149	1.142	1.131	1.121	1.110	1.104	1.100	1.095	1.092	1.086	1.081		
0.13	0.26	2.311	1.556	1.385	1.311	1.270	1.244	1.226	1.212	1.202	1.194	1.182	1.171	1.159	1.153	1.148	1.143	1.139	1.133	1.127		
0.12	0.24	2.526	1.654	1.462	1.379	1.333	1.304	1.284	1.269	1.258	1.249	1.236	1.224	1.211	1.204	1.199	1.193	1.189	1.182	1.176		
0.11	0.22	2.778	1.763	1.545	1.453	1.401	1.369	1.347	1.331	1.318	1.308	1.294	1.280	1.266	1.258	1.253	1.246	1.242	1.234	1.227	80%	90%
0.1	0.2	3.078	1.886	1.638	1.533	1.476	1.440	1.415	1.397	1.383	1.372	1.356	1.341	1.325	1.316	1.310	1.303	1.299	1.290	1.282		
0.09	0.18	3.442	2.026	1.741	1.623	1.558	1.517	1.489	1.469	1.454	1.442	1.424	1.406	1.389	1.379	1.373	1.365	1.360	1.350	1.342		
0.08	0.16	3.895	2.189	1.859	1.723	1.649	1.603	1.572	1.549	1.532	1.518	1.498	1.478	1.459	1.448	1.441	1.432	1.426	1.416	1.406		
0.075	0.15	4.165	2.282	1.924	1.778	1.699	1.650	1.617	1.592	1.574	1.559	1.538	1.517	1.497	1.485	1.477	1.468	1.462	1.451	1.441		
0.07	0.14	4.474	2.383	1.995	1.838	1.753	1.700	1.664	1.638	1.619	1.603	1.580	1.558	1.537	1.524	1.516	1.506	1.500	1.488	1.477	90%	95%
0.065	0.13	4.829	2.495	2.072	1.902	1.810	1.754	1.715	1.687	1.666	1.650	1.626	1.602	1.579	1.566	1.557	1.546	1.539	1.527	1.515		
0.06	0.12	5.242	2.620	2.156	1.971	1.873	1.812	1.770	1.740	1.718	1.700	1.674	1.649	1.624	1.610	1.600	1.589	1.582	1.568	1.556		
0.055	0.11	5.730	2.760	2.249	2.048	1.941	1.874	1.830	1.797	1.773	1.754	1.726	1.699	1.672	1.657	1.647	1.635	1.627	1.613	1.600		
0.05	0.1	6.314	2.920	2.353	2.132	2.015	1.943	1.895	1.860	1.833	1.812	1.782	1.753	1.725	1.708	1.697	1.684	1.676	1.660	1.646		
0.045	0.09	7.026	3.104	2.471	2.226	2.098	2.019	1.966	1.928	1.899	1.877	1.844	1.812	1.782	1.764	1.752	1.737	1.729	1.712	1.697	95%	
0.04	0.08	7.916	3.320	2.605	2.333	2.191	2.104	2.046	2.004	1.973	1.948	1.912	1.878	1.844	1.825	1.812	1.796	1.787	1.769	1.752		
0.035	0.07	9.058	3.578	2.763	2.456	2.297	2.201	2.136	2.090	2.055	2.028	1.989	1.951	1.914	1.893	1.879	1.862	1.852	1.832	1.814		
0.03	0.06	10.57	3.896	2.951	2.601	2.422	2.313	2.241	2.189	2.150	2.120	2.076	2.034	1.994	1.970	1.955	1.936	1.924	1.902	1.883		
0.025	0.05	12.70	4.303	3.182	2.776	2.571	2.447	2.365	2.306	2.262	2.228	2.179	2.131	2.086	2.060	2.042	2.021	2.009	1.984	1.962		
0.02	0.04	15.89	4.849	3.482	2.999	2.757	2.612	2.517	2.449	2.398	2.359	2.303	2.249	2.197	2.167	2.147	2.123	2.109	2.081	2.056	98%	
0.017	0.034	18.71	5.284	3.712	3.166	2.895	2.734	2.628	2.553	2.498	2.454	2.392	2.333	2.276	2.243	2.222	2.195	2.180	2.150	2.123		
0.015	0.03	21.21	5.643	3.896	3.298	3.003	2.829	2.715	2.634	2.574	2.527	2.461	2.397	2.336	2.301	2.278	2.250	2.234	2.202	2.173		
0.012	0.024	26.51	6.338	4.241	3.541	3.200	3.000	2.870	2.778	2.710	2.658	2.582	2.511	2.442	2.403	2.378	2.346	2.328	2.292	2.261		
0.01	0.02	31.82	6.965	4.541	3.747	3.365	3.143	2.998	2.896	2.821	2.764	2.681	2.602	2.528	2.485	2.457	2.423	2.403	2.364	2.330		
0.007	0.014	45.47	8.363	5.175	4.173	3.700	3.428	3.253	3.131	3.041	2.972	2.873	2.781	2.693	2.642	2.610	2.570	2.547	2.501	2.462	99%	99.5%
0.005	0.01	63.65	9.925	5.841	4.604	4.032	3.707	3.499	3.355	3.250	3.169	3.055	2.947	2.845	2.787	2.750	2.704	2.678	2.626	2.581		
0.003	0.006	106.1	12.85	6.994	5.321	4.570	4.152	3.887	3.705	3.573	3.472	3.330	3.197	3.073	3.003	2.957	2.902	2.870	2.808	2.754		
0.002	0.004	159.2	15.76	8.053	5.951	5.030	4.524	4.207	3.991	3.835	3.716	3.550	3.395	3.251	3.170	3.118	3.055	3.018	2.946	2.885		
0.001	0.002	318.3	22.32	10.21	7.173	5.893	5.208	4.785	4.501	4.297	4.144	3.930	3.733	3.552	3.450	3.385	3.307	3.261	3.174	3.098		
.0005	0.001	636.6	31.59	12.92	8.610	6.869	5.959	5.408	5.041	4.781	4.587	4.318	4.073	3.850	3.725	3.646	3.551	3.496	3.391	3.300	99.9%	99.95%
.0001	.0002	3183	70.70	22.20	13.03	9.678	8.025	7.063	6.442	6.010	5.694	5.263	4.880	4.539	4.352	4.234	4.094	4.014	3.862	3.733		
5e-5	.0001	6366	99.98	27.99	15.54	11.18	9.082	7.884	7.120	6.594	6.210	5.694	5.239	4.837	4.620	4.482	4.320	4.228	4.053	3.906		